KAIZEN MATHS

Lesson starter

Algebraic substitution

Quadratic expressions

4 questions · about 5 minutes

Name _____

Evaluate each expression. Use brackets for negative values.

S1 Given $x = 5$

$$x^2 + 2x$$

S2 Given $x = -3$

$$x^2 + 3x + 5$$

S3 Given $x = -4$

$$x^2 + 3x - 1$$

S4 Given $x = -1$

$$x^2 + 5x - 8$$

Show your working

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AS-02 · S · Starter

Show your working

KAIZEN MATHS

Exit ticket

Algebraic substitution

Quadratic expressions

Name _____

Evaluate each expression.

E1 Given $x = -1$

$$x^2 + 5x - 8$$

E2 Given $x = 3$

$$x^2 + 4x - 7$$

E3 Given $x = -4$

$$x^2 + 2x + 6$$

AS-02 · E · 3 questions

Show your working

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Exit ticket

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AS-02 · E · 3 questions

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AS-02 · E · 3 questions

Show your working

Teacher answers & working

AS-02 · Quadratic expressions

Page 1: two starters. Page 2: four exit tickets. Cut along the dashed lines.

Print student pages single-sided at 100%. Match S and E labels to the solutions below.

S1

Given x=5

$$x^2 + 2x$$

Substitute x=5.

$$(5)^2 + 2\times 5$$

$$25 + 10$$

$$35$$

S2

Given x=-3

$$x^2 + 3x + 5$$

Substitute x=-3.

$$(-3)^2 + 3\times(-3) + 5$$

$$9 - 9 + 5$$

$$5$$

S3

Given x=-4

$$x^2 + 3x - 1$$

Substitute x=-4.

$$(-4)^2 + 3\times(-4) - 1$$

$$16 - 12 - 1$$

$$3$$

S4 / E1

Given x=-1

$$x^2 + 5x - 8$$

Substitute x=-1.

$$(-1)^2 + 5\times(-1) - 8$$

$$1 - 5 - 8$$

$$-12$$

E2

Given x=3

$$x^2 + 4x - 7$$

Substitute x=3.

$$(3)^2 + 4\times 3 - 7$$

$$9 + 12 - 7$$

$$14$$

E3

Given x=-4

$$x^2 + 2x + 6$$

Substitute x=-4.

$$(-4)^2 + 2\times(-4) + 6$$

$$16 - 8 + 6$$

$$14$$